

UNIVERSITY OF MORATUWA

FACULTY OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

BSc Engineering Honours Degree

Semester 4 Examination: 2025

CS3023: SOFTWARE ENGINEERING

- Time allowed: 2 Hours
- ADDITIONAL MATERIAL: None
- June 2025

INSTRUCTIONS TO CANDIDATES:

1. This paper consists of 3 questions in 2 pages.
2. Answer all 3 questions.
3. Start answering each of the 3 main questions on a new page.
4. The maximum attainable mark for each question is given in brackets.
5. This examination accounts for 60% of the module assessment.
6. This is a close-book examination. **NB:** It is an offence to be in possession of unauthorised material during the examination.
7. Only calculators approved and labelled by the Faculty of Engineering are permitted.
8. Assume reasonable values for any data not given in or with the examination paper. Clearly state such assumptions made on the script.
9. In case of any doubt as to the interpretation of the wording of a question, make suitable assumptions and clearly state them on the script.
10. This paper should be answered only in English.

Question 1 (30 marks)

(i) Redefine the job role of a software engineer in the current state of the industry. Your answer must consider the software industry practices, nature of the work, technology used, professional conduct and any other relevant factor of importance [5 marks]

(ii) A given project is scoped at 50KLOCs. Using the base model of algorithmic cost estimation, compare the development scenarios of a) group of intern developers b) fully AI-integrated CASE workbench against the typical conditions. What happens to your findings when scope increases to 1MLOC? [5 marks]

(iii) Model the effort estimation variability over the standard SDLC and explain how CoCoMo2 helps to manage this challenge. [5 marks]

(iv) There is a hype in microservices as well as converting monoliths to microservices. Elaborate on the strengths of both methods (Monoliths and Microservices) for software engineering [5 marks]

(v) Describe the following:

- a. Agile vs DevOps.
- b. Containerisation
- c. AI in Spiral
- d. xUML [2.5 \times 4 \text{ marks}]

Question 2 (40 marks)

A leading private healthcare service provider in Sri Lanka plans to introduce an online hospital management solution starting with the patient management subsystem and laboratory services subsystem. Then it expects to implement the other main sub-systems such as pharmacy, finance, HR, operations theatre management, etc. The final system must comply with the standard functions of a given hospital management software. You have been recruited as a lead software engineer for the project by the hospital to develop the solution. Clearly state your assumptions when answering the following questions.

(i) Identify three key stakeholders of this system. Briefly justify your answer. [3 marks]

(ii) Draw three use case diagrams each for the identified stakeholder. Text descriptions are not required. [3 marks]

(iii) State and describe three (3) functional and non-functional requirements for the system. [6 marks]

(iv) Draw a suitable UML sequence diagram for the patient registration process. State your assumptions. [4 marks]

(v) Briefly describe the Architecture at large of the proposed system using suitable illustrations. [5 marks]

(vi) Prepare complete architecture diagrams using the following perspectives for the proposed system with sufficient finer level details.

- a. Software component level breakdown using a class diagram [5 marks]
- b. Software component-level interactions using sequence diagrams [5 marks]

(vii) Prepare a project plan for this project with key information. [5 marks]

(viii) Identify a technology and/or product stack to implement your architecture presented above; provide a sketch diagram showing how each identified item operates in the final solution. [4 marks]

Question 3 (30 marks)

(i) Your start-up company is building a Laboratory Information Management System (LIMS) to manage samples, associated data, workflows, and regulatory compliance in research and

testing labs. It is expected that this software can be used in places such as clinical and diagnostic labs, environmental testing labs, food and beverage testing labs, forensic labs as well as labs in academic and research institutes. Compare the pros and cons of implementing this LIMS as a traditional on-premises product versus offering it as a Software-as-a-Service (SaaS) solution. Your answer should consider the points-of-view of your start-up company as well as your customers (i.e. the labs). [6 marks]

(ii) Briefly explain what Infrastructure as Code is and its benefits. [4 marks]

(iii) Explain the difference between centralized and distributed code management systems. (You may use diagrams to aid your explanation). Give examples of tools which represent each type of code management system. [8 marks]

(iv) List the components that are commonly found in a Real-time Operating System (RTOS) and briefly explain what each one means? [8 marks]

(v) In an RTOS, at anytime there can be several processes which are eligible for execution. Briefly explain the two main strategies used for scheduling processes in an RTOS. [4 marks]

END OF THE PAPER